SWACHHMAP

AI-Powered Smart Waste Reporting & Management Platform

Tagline: Speak Up. Report Smart. Build a Cleaner Tomorrow.

Executive Overview & Vision

SwachhMap is an AI-powered civic technology platform designed to transform waste reporting into an effortless, transparent, and conversational process. Guided by the vision to make reporting as simple as having a conversation, SwachhMap empowers citizens to actively participate in building cleaner, more sustainable communities.

Core Value Proposition

The Problem

Traditional waste reporting systems rely on complex, cumbersome forms, error-prone manual location entries, and offer little to no transparency into complaint resolution. This friction discourages civic participation while leaving municipal authorities without a unified, real-time mechanism to monitor and resolve issues.

The Solution: Conversational & Vision AI

SwachhMap replaces clunky interfaces with **Swachh Saathi**, an intelligent multilingual voice assistant that guides users naturally through reporting. Coupled with computer vision and automated geolocation, reporting requires minimal user effort while delivering verified, actionable data to city administrators.

Key Capabilities & Architecture

Core Features

- **Multilingual AI Voice Assistant (Swachh Saathi):** Enables natural voice interaction across regional languages to log complaints conversationally.
- **Computer Vision Verification (YOLO11n):** Automatically scans and verifies garbage in uploaded visual evidence to ensure legitimacy.
- **Smart Geolocation & Audio Confirmation:** Pinpoints exact GPS coordinates, reverse geocodes them into human-readable locality names, and speaks the location aloud for user confirmation.

- **Interactive Mapping:** Powered by Leaflet & OpenStreetMap for real-time visualization of reported waste spots.
- **Citizen Tracking & Admin Dashboard:** Provides citizens with end-to-end status visibility via 'My Reports', while equipping administrators with actionable resolution workflows.

Technology Stack

Layer	Technologies
Frontend	React, JavaScript
Backend & Database	Node.js, Express.js, MongoDB
Conversational AI	Gnani AI (Prisma STT, Evon LLM, Timbre TTS)
Computer Vision	YOLO11n
Mapping & Geospatial	Leaflet, OpenStreetMap, GPS, Reverse Geocoding
Media Storage	Cloudinary

End-to-End Workflow

Speak → Understand → Detect → Locate → Verify → Submit → Track

1. **Speak & Understand:** Citizen speaks the issue naturally; Swachh Saathi processes speech and intent.
2. **Detect & Locate:** YOLO11n analyzes the photo while GPS identifies the exact coordinates and announces the locality.
3. **Verify & Submit:** User confirms the detected details and submits with a single prompt.
4. **Track & Resolve:** Citizen monitors real-time resolution; administrative dashboards streamline cleanup logistics.

Innovation & Civic Impact

By unifying conversational AI, computer vision, and geospatial mapping, SwachhMap eliminates barriers of literacy and technical complexity. It fosters municipal accountability, dramatically accelerates cleanup response times, and provides a scalable blueprint for smarter, cleaner cities.